10 years of ELIXIR CZ

elixir CZECH REPUBLIC

The national infrastructure for biological data

Over the past decade, ELIXIR CZ has significantly advanced life science research by organizing, storing, sharing, and facilitating data interoperability. As a key member of the pan-European infrastructure, we've supported the national and European scientific community through **strategic areas** including **Chemical Biology**, **Structural Bioinformatics**, **Genomics**, **Human Data, Data Management and Access Control**. Our comprehensive training programs and suite of tools, like IDSM, Protein Engineering Portal, RepeatExplorer, Life Science AAI, and Data Stewardship Wizard have empowered researchers to achieve groundbreaking insights and innovations.

Personalized Medicine Enabled by Data

ELIXIR CZ leverages high-quality, accessible data to push the boundaries of personalized medicine. By analyzing genetic variations and clinical data, ELIXIR CZ helps medical professionals design tailored treatments, particularly for complex diseases like cancer and neurodegenerative disorders. This approach not only enhances the precision of treatments but also exemplifies the transformative potential of data-driven healthcare.

A Gateway to European Data

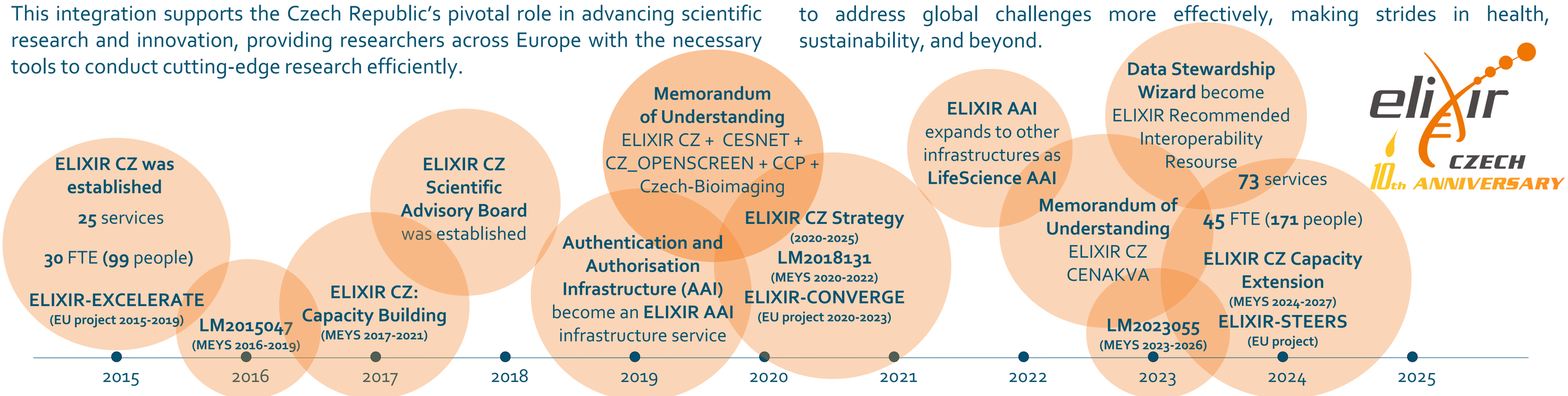
As a central node in the European data infrastructure, ELIXIR CZ facilitates seamless access to vast biological datasets, tools, and computational resources. This integration supports the Czech Republic's pivotal role in advancing scientific research and innovation, providing researchers across Europe with the necessary tools to conduct cutting-edge research efficiently.

AI's Role in Refining Data Processes

Artificial Intelligence (AI) plays a crucial role in analyzing the vast amounts of data managed by ELIXIR CZ. From predicting protein structures to discovering new genetic correlations, AI enhances the capability to extract meaningful insights from complex datasets, ensuring that data not only supports current research but also paves the way for novel discoveries in genomics and biomedicine.

The Future of Data and AI in Science

The future of scientific research with ELIXIR CZ at the helm is marked by deeper integration of AI and an expanded European infrastructure, promoting open science across borders. This evolution will further enable the scientific community to address global challenges more effectively, making strides in health, sustainability, and beyond.



How can ELIXIR help you with your research?

The scientific and technical activities in ELIXIR are run by **5 Platforms** and **18 Communities (CZ** participates in **10)** and **10 Focus Groups (8 CZ)**.
Platforms bring together experts to define the strategy and provide services in a particular area.

To manage the huge increase in life science data

Compute Platform enables researchers across Europe to use existing facilities to access, store, transfer and analyse large and complex datasets in the cloud.
Data Platform establishes quality criteria for data resources.

To deal with the increasing complexity of data

Interoperability Platform establishes standards that can be used to describe life science data, thereby making different datasets easier to compare and analyse together.

To provide an opportunity to be a part of a wide-reaching network

Communities bring together scientists who work in a specific domain. These scientists develop services targeted for their own domains. They also give feedback on the services the Platforms provide, to make sure that the services solve real world problems.

To make it easier to find the right tools and training

Tools Platform benchmarks tools for quality to help researchers choose the most suitable tool for their research.
Training Platform enables researchers to learn how to use these tools, as well as increases the discoverability of a range of training resources through a registry covering all aspects of bioinformatics and computational biology.

To enable collaboration and skill development

Staff Exchange supports the exchange of staff between ELIXIR Nodes for around 3-6 months.
Travel grant provides funding to attend ELIXIR-related events or to travel for short-term visits to another Node.
Knowledge Exchange allows ELIXIR Node members to spend a short period of time working at industry partners' facilities.

ELIXIR's impact

Bioinformatics resource uptake: we work to increase resource usage and appreciation by users

Benefits derived from working together: (i.e. relationship/social capital), we facilitate knowledge-sharing and cooperation

Research efficiency: we make infrastructure, bioinformatics resources and processes faster, easier to use, and more integrated

Scientific legacy: we create and disseminate new knowledge on research infrastructure, bioinformatics resources and related guidelines

Skills development: (i.e. human capital), we upskill resource users and service providers

Equal opportunity: we raise awareness of diversity and inclusiveness

Contacts

Jiří Vondrášek - Head of ELIXIR CZ
jiri.vondrasek@uochb.cas.cz

Anna Strachotová
anna.strachotova@uochb.cas.cz

https://www.elixir-czech.cz

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